YZM5110 Multimodal Learning, research project

Reproducing and Speeding Up Visual Contrastive Decoding on POPE

A replication of Leng et al. (CVPR 2024) on LLaVA-1.5-7B, and a check that skips the second pass of VCD without changing any answer

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19 November 2026



Models say “yes” to objects that are not there



My drawing, not a COCO image. There is no fork in it.

6%

of absent objects got a “yes” in my runs on the easiest POPE split (random)

14%

on the hardest split (adversarial)

LLaVA-1.5-7B, regular decoding, mean of 3 seeds. Each split asks 1,500 questions about absent objects.

POPE turns hallucination into yes/no questions



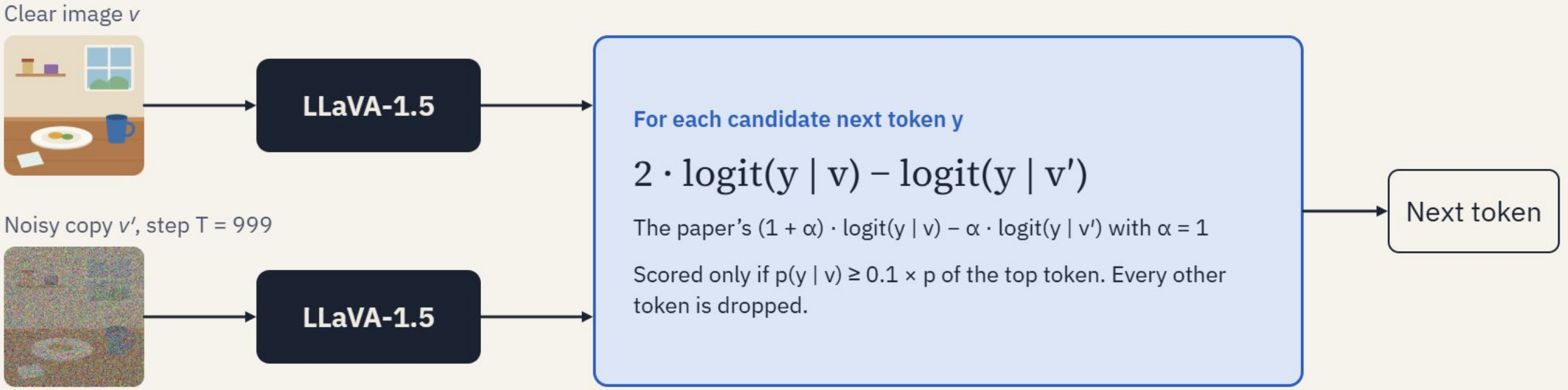
Same drawing, asked in POPE's format. The real test uses 500 COCO images with three present and three absent objects each: 3,000 questions per split, half of them "yes".

Every question reads "Is there a [object] in the image?"

Object	Why POPE asks about it	Answer
cup	it is in the image	yes
dining table	it is in the image	yes
giraffe	random pick (random split)	no
person	most common in COCO (popular split)	no
fork	often next to a table (adversarial split)	no

F1 counts "yes" as the positive answer. A model that says "yes" more often finds more of the real objects and can raise F1 without seeing any better. So I show the share of "yes" answers next to every F1.

VCD compares a clear image with a noisy copy



> Paper, Table 1: +5.8, +5.0 and +3.8 F1 over regular decoding on the three splits. No training.

Leng et al., CVPR 2024: method in Section 3, POPE results in Table 1

VCD reproduces only at the paper's noise level



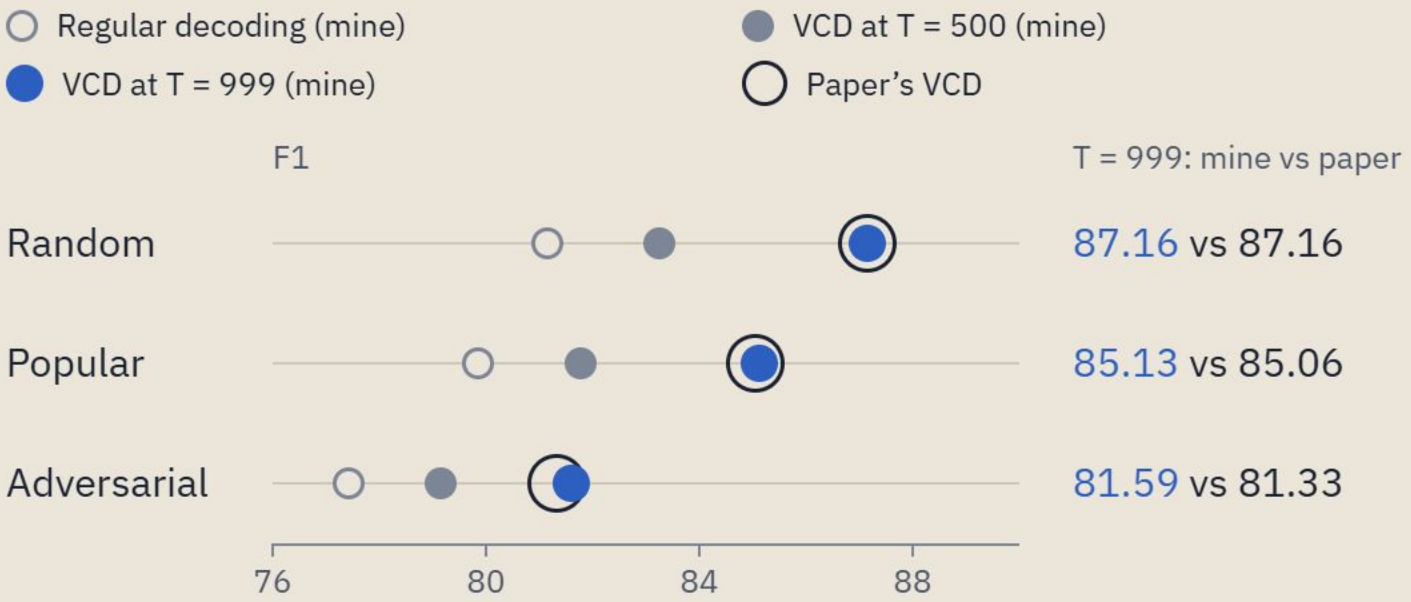
Clear image
No noise added



T = 500
86% of the signal left, by my calculation. The value in the official POPE script



T = 999
28.5% left, by the same calculation. The value in the paper's appendix



In my runs VCD at T = 999 also says "yes" more often: 46 to 53% of answers, against 39 to 43% with regular decoding. Part of the F1 gain comes from that shift.

F1 on POPE MSCOCO. Mine: mean of 3 seeds; at T = 999, 2 seeds (popular: 1). Paper: Leng et al., Table 1.

The noisy pass is wasted when one token is left

1 Run the model on the clear image

↓

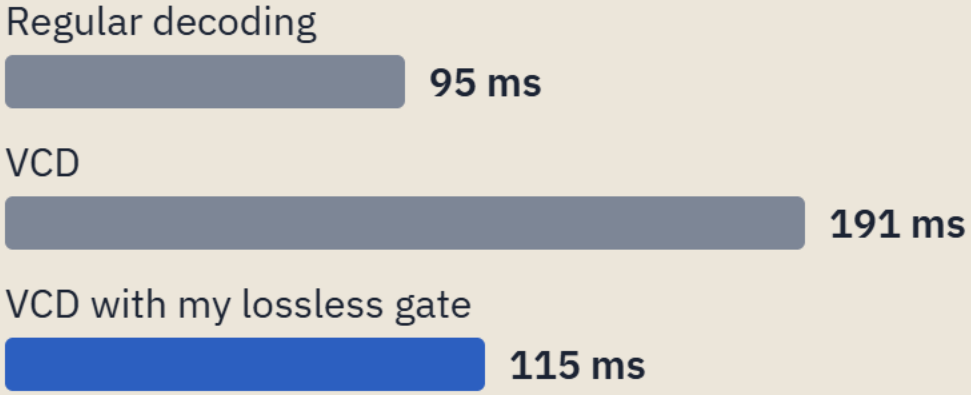
2 Keep every token whose probability is at least 0.1 × the top one

↓ ↓

One token left: output it and skip the noisy pass

Two or more: run the noisy pass as VCD does

Time per question, 1,000 COCO random questions



9,000 of 9,000

answers identical to VCD, all three splits

61 to 71%

of questions in my runs never needed the noisy pass

One A100 80 GB with no other job running. Answer check: seed 1 on all three splits. VCD at T = 500.

If you use POPE

Report the share of “yes” answers next to F1.

VCD matches the paper at noise step 999, not at the 500 in the official script. Skipping the noisy pass when one token is left kept all 9,000 answers and cut the time per question from 191 to 115 ms.

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Thank you